Technical Reference Manual: IESL Robogames – Custom Line Following Robot

This document provides a comprehensive technical breakdown of the IESL Robogames custom Line Following Robot.

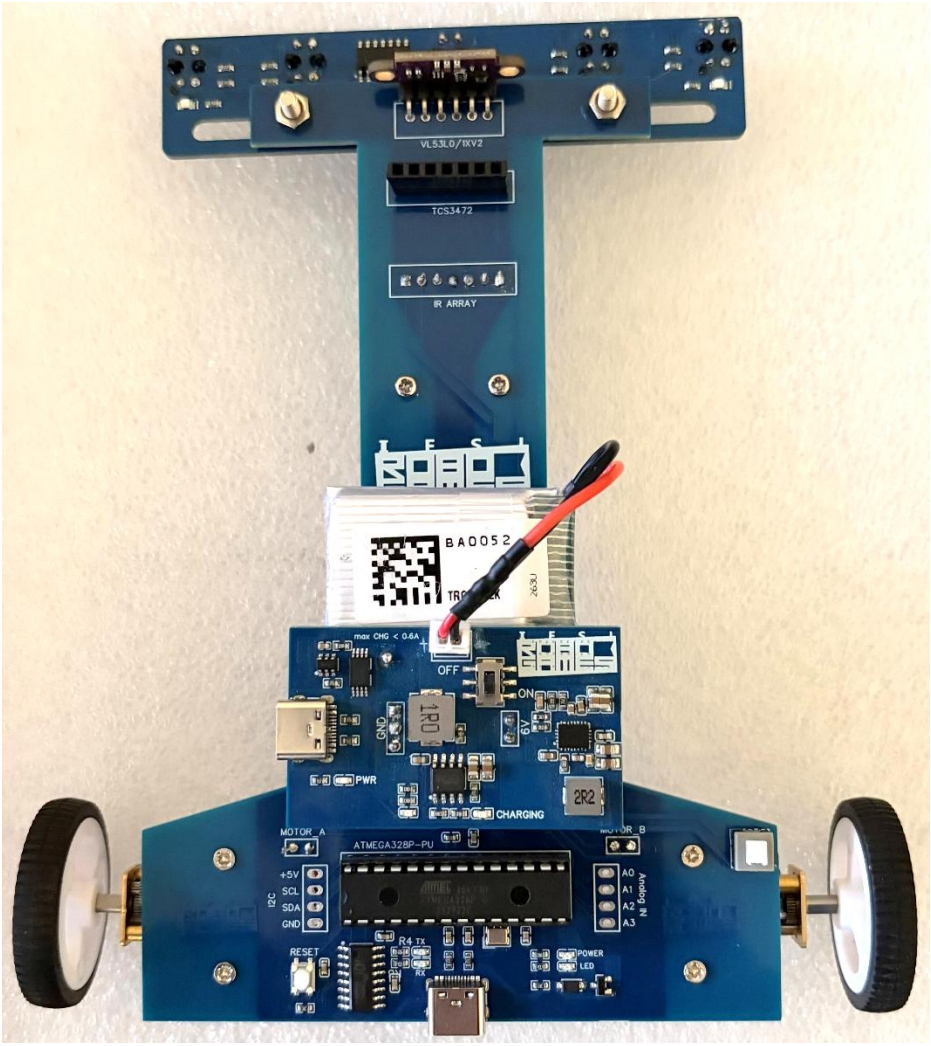

Figure 1

1. Hardware Specifications

Microcontroller: The core of this robot is the **ATmega328P**. It is the same microcontroller found in the **Arduino Uno**, meaning all standard Arduino libraries and programming logic are fully compatible with this robot.

The large chip located in the center of the Robot (see **Figure 2**) is the ATmega328P. It executes the code you write and controls all sensors and motors.

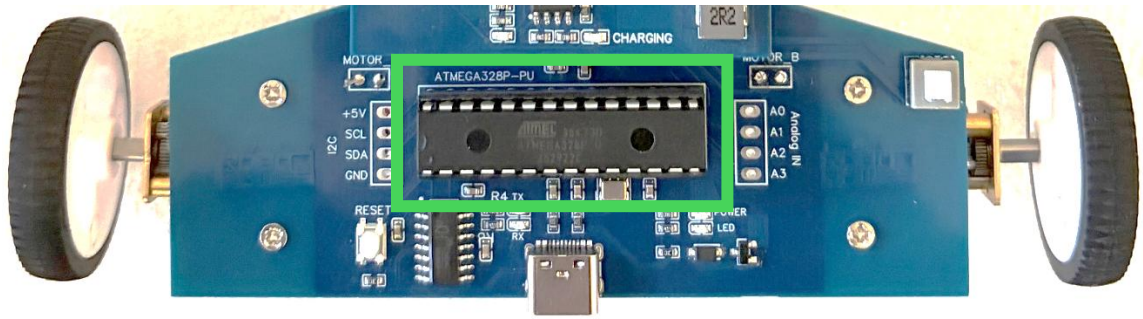

Figure 2

Programming Port The robot features a **USB-C port** located on the bottom of the chassis (see **Figure 3**). This port is the one you use for uploading the Arduino code and read Serial Monitor data

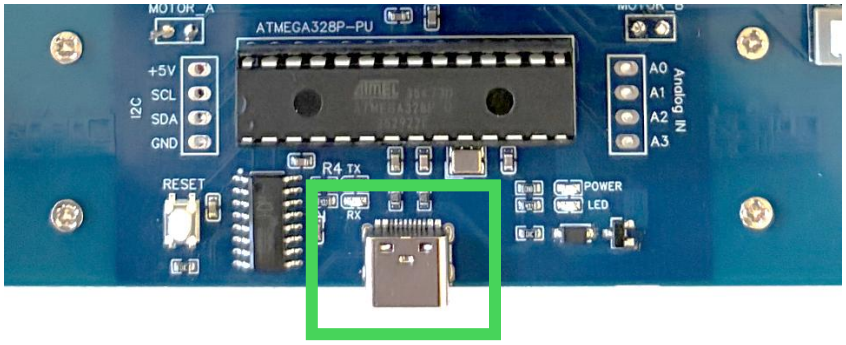

Figure 3

Power LED: Indicates the 5V power to the Robot

Built-in LED (LED): Connected to Digital Pin 13; useful for "Blink" tests and simple status signals.

TX/RX LEDs: These flash rapidly during code uploads and stays on when the robot is sending data to the Serial Monitor

Robot can be powered using battery module from the push button on the right corner (see **Figure 4**)

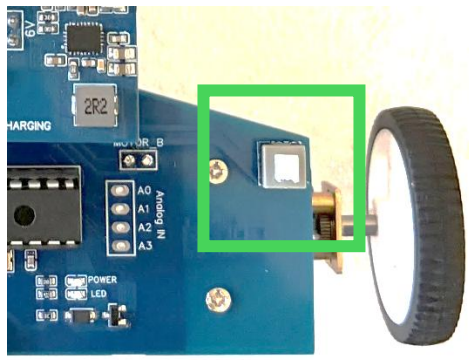

Figure 4

2. Power Module

Figure 6

The Power Module is a separate board connected to the main robot via pin headers. It is responsible for managing the **3.7V Lithium-Polymer (Li-Po) battery**, regulating the voltage and charging the battery.

Voltage Regulation The module takes the nominal 3.7V from the battery and boosts it to a stable **6V DC output**. This higher voltage is used to provide the N20 motors consistent power.

Connection (CRITICAL) The module connects to the robot via a specific pin header set. Incorrect orientation will cause permanent damage to the electronics.

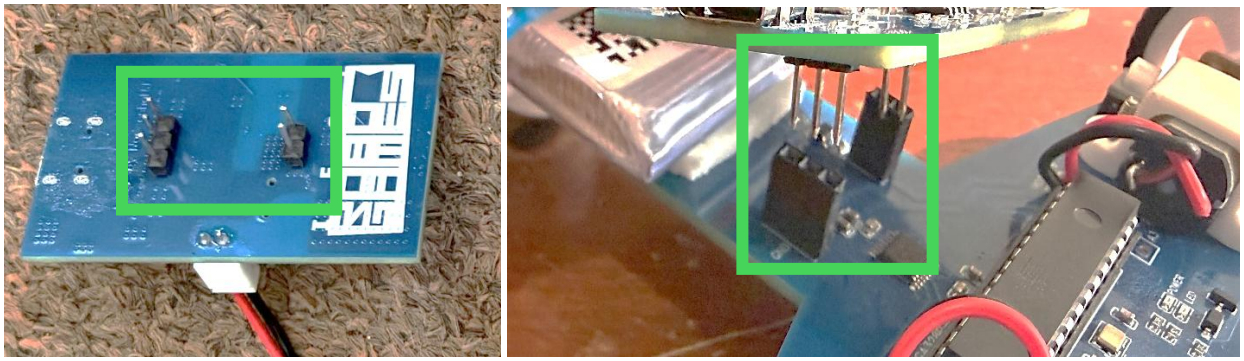

Figure 7

- **3-Pin Side:** Must be connected to the **Ground (GND)** pins on the robot.
- **2-Pin Side:** Must be connected to the **6V** pins on the robot.
- **Warning:** Do not swap or reverse this connection. Ensure the pins are aligned perfectly before pressing the module down.

Charging the battery is done via the **USB-C port located on the power module** (separate from the microcontroller's programming port). This port is only for charging and cannot be used to power the robot or program the microcontroller.

- **LED Indicators (USB Power Only):**

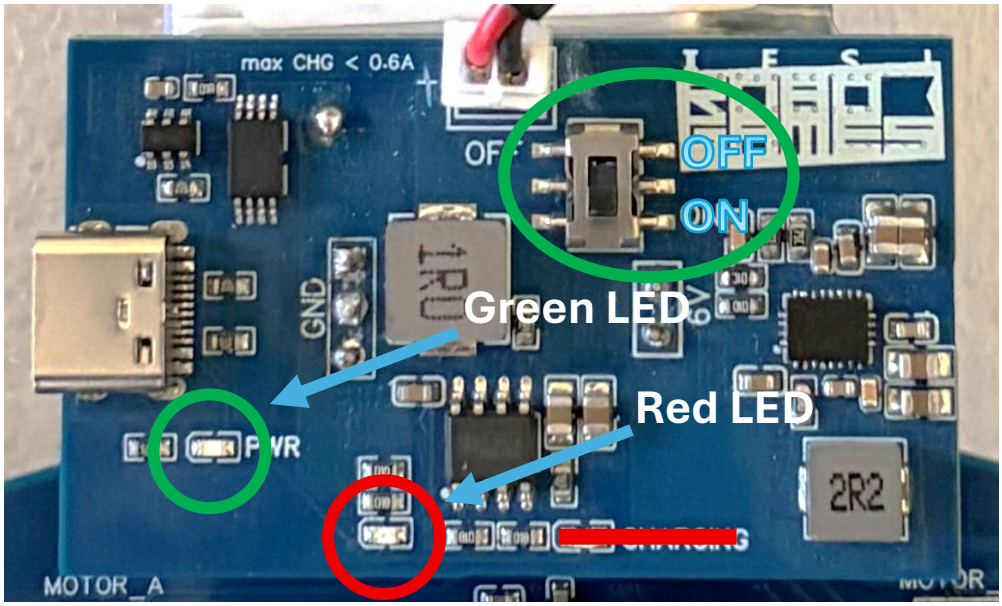

Figure 8

- **Green LED:** Indicates the module is receiving power from the USB-C source.
- **Red LED:** Indicates the battery is currently charging.(see **Figure 8)**

- **Note:** These LEDs only function when the USB-C cable is plugged in. They will not light up when the robot is running on battery power.
- **Safety Timer:** The charging circuit does not automatically cut off. **You must manually disconnect the USB-C cable after 30 minutes of charging** to prevent overcharging.

Operation & Handling

- **Main Power Switch:** To provide the 6V output to the robot, pull the switch **DOWN** to the "ON" position(see **Figure 8**) . Always switch the power module **OFF** after use to preserve battery health.
- **Battery plug:** When connecting the battery JST jack to the module, ensure the orientation matches the original configuration.

3. **Line Detection: 5-Channel IR Sensor Array**

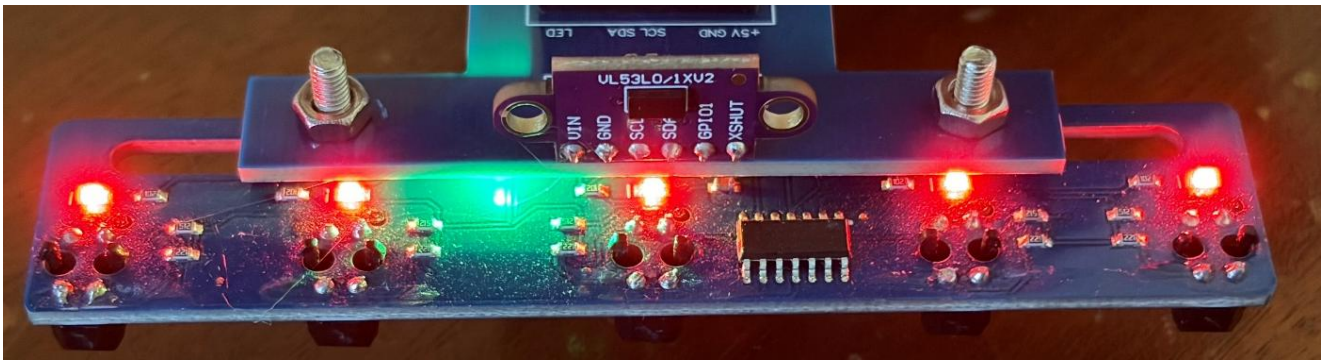

Figure 9

The robot navigates the track using a 5-channel Infrared (IR) sensor array mounted at the front. These sensors detect the contrast between the black line and the white floor.

These sensors provide **Digital Values**. In your code, you will read these as simple "on" or "off" signals to determine the black line.

Hardware Pin Configuration: To program the robot, you must reference the correct pins in your code. The sensors are mapped to the microcontroller as follows:

Sensor Position	Arduino Pin
Far Left	**4**
Mid Left	**3**
Center	**2**
Mid Right	**7**
Far Right	**8**

Note: Order of these pins are subjected to change depends on the wiring

Mechanical Adjustment The sensitivity and accuracy of the line following depend on the distance between the sensors and the floor.

- **Height Tuning:** You can adjust the sensor height by loosening or tightening the **mounting bolt** that secures the array to the robot chassis.
- **Optimal Range:** Generally, keeping the array 3mm–5mm above the ground provides the most stable readings.

Maintenance & Handling The connection between the sensor array and the main robot body consists of delicate **soldered wiring**. Do not pull, twist or put tension on these wires.

4. Distance Sensor (VL53L0X)

The robot uses a **VL53L0X Time-of-Flight (ToF)** sensor to measure distance.
This module communicated via I2C protocol using hardware address 0x29. For programming, you can use the **Adafruit_VL53L0X** library available in the Arduino IDE. Additionally, the sensor's **XSHUT pin** is connected to **Digital Pin 11** can be used to change hardware I2C address or put it into a low-power standby mode.

Figure 10

5. Motor Drive System

The robot uses two **N20 Gear Motors (300 RPM)** for movement. These motors are driven by a **DRV8833 dual motor driver** integrated into the board.

Pin Configuration:
The motors are connected to the microcontroller via PWM-enabled pins.
Left motor is controlled using Arduino **Digital Pins 5,6** and Right motor is controlled using Arduino **Digital Pins 9,10**

Initial Throttle: If the robot appears to "stutter" or get stuck immediately when powered on, try reducing your starting speed/throttle value in the code to allow the gears to engage smoothly.

6. Maintenance & Support

Battery Health To maximize the lifespan of the Lithium-Polymer battery, follow these critical care guidelines:

- **Voltage Limit:** Never allow the battery voltage to drop below **3.0V**, as this will permanently reduce battery health and capacity.
- **Storage:** Do not store the robot for long periods with a completely drained battery.
- **Usage:** Avoid using the robot for extended periods without recharging. If the motors seem sluggish, stop and charge the module immediately.

Developer Support This robot was designed to be a flexible platform for learning. If you encounter hardware behaviors or technical issues that cannot be solved by the troubleshooting guide in this manual, please consult the **Technical Support team of the IESL RoboGames**.

Good luck at the IESL RoboGames!